



Weather Trend Forecasting — Project Report

PM Accelerator Technical Assessment Dataset: Global Weather Repository (Kaggle) — daily weather for cities worldwide, 40+ features Author run: 149,880 records · 211 countries · 268 cities · 2024-05-16 → 2026-06-27

 PM Accelerator Mission. The Product Manager Accelerator Program supports PM professionals through every stage of their careers — from students seeking entry-level roles to Directors stepping into leadership — via certified training programs, hands-on projects, coaching and career services, while fostering a diverse landscape in the tech industry. Through its PMA Kids initiative it also offers free Product Management education to teenagers from underserved families. <https://www.pmaccelerator.io/>

This report presents all analyses, model evaluations and visualizations. An interactive version of everything below is also available as a Streamlit dashboard (streamlit run dashboard.py).

Requirements Checklist

#	Requirement	Status	Where (section / figure)
Basic	Handle missing values, outliers, normalize		§1 Data Cleaning
	EDA — trends, correlations, patterns		§2 EDA
	Visualizations for temperature & precipitation		§2 — Figs 1–3
	Forecasting model + evaluation metrics		§3 — Table 1, Figs 6–7

#	Requirement	Status	Where (section / figure)
	Use last_updated for time-series analysis	✓	§3 (daily series indexed by last_updated)
Advanced	Anomaly detection & outlier analysis	✓	§4 — Fig 9
	Build & compare multiple forecasting models	✓	§3 — Table 1 (5 models)
	Ensemble of models	✓	§3 — "Ensemble" row
	Climate analysis (long-term, regional)	✓	§5 — Fig 10
	Environmental impact (air quality vs weather)	✓	§6 — Fig 11
	Feature importance	✓	§7 — Fig 12
	Spatial analysis (geographical viz)	✓	§8 — Fig 13
	Geographical patterns (countries/continents)	✓	§9 — Figs 5, 14
Deliverables	PM Accelerator mission displayed	✓	Top of report + dashboard
	Report with analyses/evaluations/visuals	✓	This document
	README + requirements file	✓	README.md, requirements.txt

Basic Assessment

1. Data Cleaning & Preprocessing

The dataset has one record per city per update, with last_updated as the timestamp and 40+ numeric/categorical features. Cleaning steps (src/data_cleaning.py):

1. Type parsing — last_updated parsed to datetime for time-series use.
2. Duplicates — exact duplicates and duplicate (location, last_updated) pairs removed (kept latest).
3. Missing values — rows without a timestamp/temperature target dropped; numeric columns median-imputed (robust to skew), categorical mode-imputed. *(The raw file is already complete — 0 missing — so cleaning mainly adds robustness and structure.)*
4. Outliers — core variables winsorized to the Tukey IQR fence ($Q1 - 1.5 \cdot IQR$, $Q3 + 1.5 \cdot IQR$) — clipping keeps every record while neutralising sensor spikes.
5. Normalization — z-score standardized copies (*_z) added for modelling; originals kept for interpretable plots.
6. Feature engineering — calendar features (year, month, day, dayofyear, hour, season) derived.

Result: 149,880 → 149,879 clean rows · 0 missing · 0 duplicates.

2. Exploratory Data Analysis

Temperature

Temperature is near-symmetric; clear seasonal structure appears in the per-season boxplots.

Temperature distribution and seasonality *Fig 1 — Temperature distribution (°C) and spread by season.*

Precipitation

Precipitation is highly right-skewed (most readings near zero, long tail); the monthly mean reveals the wet/dry cycle.

Precipitation overview *Fig 2 — Precipitation distribution (log scale) and mean precipitation by month.*

Global trend over time (last_updated)

Global temperature & precipitation trend *Fig 3 — Daily global-mean temperature and precipitation across the record.*

Correlations

Temperature tracks feels-like almost perfectly ($r = 0.98$) and rises with UV (+0.49), while pressure (-0.44) and humidity (-0.35) pull it down; precipitation is nearly uncorrelated with temperature (+0.05).

Correlation heatmap *Fig 4 — Pearson correlation matrix of the main weather variables.*

Country patterns

Coollest mean-temperature countries: Iceland (5.7 °C), Canada (7.5 °C), Norway (8.5 °C).
Hottest: Saudi Arabia ≈ 45 °C, Morocco ≈ 40 °C.

Country temperature ranking *Fig 5 — Hottest and coolest countries by mean temperature.*

3. Model Building — Forecasting & Evaluation

Target. All cities are aggregated to a single daily global-mean temperature series indexed by last_updated, gap-filled by time interpolation (src/forecasting.py).

Models compared (Basic + Advanced "multiple models" + ensemble):

Model	Type	Notes
Naive (lag-1)	Baseline	Yesterday's value
Linear Regression	ML	Lag + rolling + cyclical calendar features
Random Forest	ML	Non-linear, same features

Model	Type	Notes
SARIMAX	Statistical	order=(3,1,1)
Ensemble	Combination	Mean of Linear + RF + SARIMAX

Evaluation. Held-out last 30 days, 1-step-ahead walk-forward for every model (a fair, like-for-like comparison). Metrics: MAE, RMSE, MAPE, R^2 .

Table 1 — Model evaluation (held-out test):

Model	MAE (°C)	RMSE (°C)	MAPE (%)	R^2
Naive (lag-1)	0.190	0.242	0.83	0.767
Ensemble	0.212	0.249	0.93	0.752
SARIMAX	0.193	0.265	0.84	0.719
Random Forest	0.235	0.274	1.03	0.701
Linear Regression	0.237	0.281	1.04	0.686

Forecast comparison *Fig 6* — Each model's predictions vs. actuals on the held-out window.

Model error comparison *Fig 7* — MAE / RMSE by model (lower is better).

Reading the result. Because the target is a *planetary daily mean* over 268 cities, it is extremely smooth day-to-day, so the lag-1 baseline is a strong competitor ($R^2 \approx 0.77$). SARIMAX and the ensemble track within ~ 0.02 RMSE — an honest, common finding for highly-aggregated series. The best statistical model, refit on all data, produces the 30-day forecast below (avg ≈ 23.8 °C).

30-day forecast *Fig 8* — 30-day-ahead global mean temperature forecast (refit on all data).

Advanced Assessment

4. Anomaly Detection

An IsolationForest over temperature, precipitation, humidity, wind and pressure flags 2,998 of 149,879 records (2.0%) as multivariate anomalies (unusual *combinations* of conditions). The dashboard additionally cross-checks with Z-score and IQR univariate methods.

Anomaly detection *Fig 9 — Anomalies (red) vs. normal records in temperature–humidity space.*

5. Climate Analysis

Monthly climatology (mean $\pm$ std) plus seasonality split by latitude band shows tropical zones staying warm year-round while temperate/polar zones swing strongly — and the two hemispheres move in opposite phase.

Climate analysis *Fig 10 — Monthly temperature climatology and seasonality by latitude band.*

6. Environmental Impact — Air Quality

Correlating pollutants (CO, O₃, NO₂, SO₂, PM2.5, PM10, EPA/DEFRA indices) with weather: the strongest links are humidity $\leftrightarrow$ ozone (−0.40) and UV $\leftrightarrow$ ozone (+0.35) — consistent with photochemical ozone formation in dry, sunny conditions — with humidity anti-correlating with the air-quality indices.

Air quality correlations *Fig 11 — Air-quality pollutants vs. weather parameters.*

7. Feature Importance

Permutation importance from a Random Forest predicting temperature is dominated by latitude (0.83), then UV index (0.29), day-of-year (0.26) and pressure (0.19) — i.e. *where* and *when* explain most of the variation. (The dashboard also shows Gini and |correlation| views.)

Feature importance *Fig 12 — Permutation feature importance for temperature.*

8. Spatial Analysis

Plotting each city's latest reading on the globe traces the continents — hot equatorial/desert regions in red, cool high latitudes in blue.

Spatial temperature map *Fig 13 — Global temperature distribution (latest reading per city).*

9. Geographical Patterns

Temperature spread and mean precipitation differ sharply across the most-sampled countries, reflecting their climate zones.

Geographical patterns *Fig 14 — Temperature spread and mean precipitation by country.*

10. Key Insights & Conclusions

- Persistence dominates the global daily-mean series: a lag-1 baseline hits $R^2 \approx 0.77$; SARIMAX/ensemble track closely (≈ 0.72 – 0.75).
- Geography is the master variable — latitude alone is the top temperature driver (importance 0.83), with mirror-image seasonality across hemispheres.
- Weather shapes air quality — humidity is the strongest pollutant correlate (ozone -0.40), and sun/UV raises ozone ($+0.35$).
- The raw data is already complete (0 missing), so cleaning centres on de-duplication, IQR outlier capping and feature engineering.
- Anomaly detection isolates 2.0% of records for data-quality / extreme-event review.

11. Reproducibility

```
pip install -r requirements.txt
```

```
python -m src.download_data
```

```
python main.py
```

```
streamlit run dashboard.py
```

All figures, metrics (outputs/model_metrics.csv), the forecast (outputs/future_forecast.csv) and a summary (outputs/summary.json) are regenerated deterministically (RANDOM_STATE = 42). See ../README.md for full instructions.